# Architectural Evolution of Global Communications: A Comprehensive Study of Classical Network Taxonomies, Programmable IPv6 Fabrics, IMT-2030 (6G), Post-Quantum Cryptography, and the Quantum Internet

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## Abstract

Global telecommunications and data networking infrastructures are undergoing a foundational architectural transformation driven by the convergence of native IPv6 network programmability, disaggregated wireless control planes, ultra-low-latency storage fabrics, post-quantum cryptographic migration, and quantum mechanical information transport. Classical networking paradigms, historically reliant on rigid protocol stacks and stateful multiprotocol signaling, are increasingly unable to satisfy the deterministic latency, extreme connection density, and cryptographic resilience demanded by distributed artificial intelligence, immersive mixed reality, and autonomous industrial systems. This paper presents a comprehensive, cross-disciplinary architectural analysis of contemporary network taxonomies and their evolutionary trajectory toward 2030 and beyond. At the classical core and transport layers, the transition from Multiprotocol Label Switching (MPLS) to Segment Routing over IPv6 (SRv6, RFC 8986) embeds stateless network instructions directly within 128-bit IPv6 addresses, unifying traffic engineering, Virtual Private Networks (VPNs), and Service Function Chaining (SFC) under a single forwarding plane. Across access and data center domains, Wireless Local Area Networks (WLANs) are advancing through IEEE 802.11be (Wi-Fi 7) Extremely High Throughput (EHT) specifications, Storage Area Networks (SANs) are replacing legacy SCSI-over-Fibre Channel translation with Non-Volatile Memory Express over Fabrics (NVMe-oF) utilizing Remote Direct Memory Access (RDMA), and cellular Radio Access Networks (RANs) are disaggregating via Open RAN (O-RAN) multi-timescale intelligent controllers. Looking toward the next decade, the International Telecommunication Union (ITU-R) IMT-2030 framework for

sixth-generation (6G) mobile systems establishes stringent performance boundaries—including 0.1 to 1ms air-interface latency, $10^6$ to $10^8$ devices/km$^2$ connection density, and centimeter-level positioning—while integrating sensing and AI natively into the network fabric. Concurrently, the existential threat posed by quantum algorithms to classical asymmetric cryptography has catalyzed the standardization of Module-Lattice and Stateless Hash-Based Post-Quantum Cryptography (NIST FIPS 203, FIPS 204, and FIPS 205), requiring complex enterprise-wide protocol migrations. Ultimately, the emergence of the Quantum Internet—operating via quantum superposition, multipartite entanglement, quantum repeaters, and quantum teleportation—establishes an entirely new physical and logical communication paradigm capable of information-theoretic security and distributed quantum computation toward the 7G horizon.

**Keywords:** Segment Routing over IPv6 (SRv6), RFC 8986, Wi-Fi 7 (IEEE 802.11be), NVMe over Fabrics (NVMe-oF), Open RAN (O-RAN), IMT-2030 (6G), Integrated Sensing and Communication (ISAC), Post-Quantum Cryptography (PQC), NIST FIPS 203 (ML-KEM), Quantum Internet, Quantum Repeaters, Entanglement Swapping.

## 1. Foundational Network Architectures and Taxonomy of Contemporary Networks

Modern communication networks are structured around the logical separation of operational planes—specifically the Forwarding Plane (or Data Plane), Control Plane, Management Plane, and Application Plane, as formally codified by the Internet Engineering Task Force (IETF) in RFC 7426. In classical digital communication systems, the Forwarding Plane transports discrete logical states represented as binary bits ("1" or "0") encoded into electromagnetic or optical signals across physical media. While the fundamental objective of state transfer remains constant, the architectural realization of a network diverges sharply based on spatial coverage, latency tolerance, state maintenance, and medium access control.

Rather than viewing network types merely as geographical classifications—such as Local Area Networks (LANs), Storage Area Networks (SANs), Radio Access Networks (RANs), and Wide Area Networks (WANs)—modern systems engineering evaluates network taxonomies by how each domain manages protocol overhead, state distribution, and hardware disaggregation. In legacy architectures, each network tier introduced specialized encapsulation and signaling protocols: Ethernet and IEEE 802.11 for local access, Fibre Channel with Small Computer System Interface (SCSI) translation for storage fabrics, proprietary monolithic basebands for cellular access, and Multiprotocol Label Switching (MPLS) coupled with auxiliary label distribution protocols for wide-area transport. This protocol fragmentation created severe operational bottlenecks, as border nodes were forced to perform computationally expensive protocol translations, maintain massive per-flow state tables, and manage siloed management planes.

Contemporary network engineering resolves this fragmentation through architectural convergence and programmability. In the local wireless domain, IEEE 802.11be (Wi-Fi 7) transforms unlicensed wireless access from a best-effort medium into a high-capacity, deterministic fabric through Multi-Link Operation (MLO). In data center storage environments, NVMe over Fabrics (NVMe-oF) eliminates the SCSI-to-NVMe translation bottleneck by extending native Non-Volatile Memory Express command sets directly across RDMA and Fibre Channel fabrics. In cellular access, the O-RAN architecture replaces monolithic base stations with disaggregated functional units governed by standardized open interfaces and closed-loop machine learning controllers. Finally, at the wide-area and service provider core, Segment Routing over IPv6 (SRv6) collapses the complex MPLS signaling stack into native IPv6 header extensions, enabling end-to-end source routing and network slicing across multi-domain topologies.

**Table 1: Structural Taxonomy and Performance Characteristics of Contemporary Network Architectures**

| Network Domain & Classification | Primary Standard / Framework | Core Transport & Signaling Mechanism | Key Architectural Innovation | Primary Operational Scope & Target Latency / Scale |
|---|---|---|---|---|
| **Wireless Local Area Network (WLAN)** | IEEE 802.11be-2024 (Wi-Fi 7 / EHT) | Multi-Link Operation (MLO) across unlicensed spectrum bands | Simultaneous multi-band aggregation and deterministic low-latency scheduling | High-density indoor/enterprise access; ultra-low local latency |
| **Storage Area Network (SAN) & Data Center Fabric** | NVMe over Fabrics (NVMe-oF) | Native NVMe over RDMA (RoCE), Fibre Channel (FC-NVMe), or TCP | Elimination of legacy SCSI-to-NVMe command translation overhead | Disaggregated data center storage; microsecond-scale fabric latency |

| Network Domain & Classification | Primary Standard / Framework | Core Transport & Signaling Mechanism | Key Architectural Innovation | Primary Operational Scope & Target Latency / Scale |
|---|---|---|---|---|
| **Wide Area Network (WAN) & Core Transport** | IETF RFC 8402, RFC 8754, RFC 8986 (SRv6) | 128-bit IPv6 Segment Identifiers (SIDs) and Segment Routing Header (SRH) | Stateless source routing; removal of LDP and RSVP-TE signaling | Global carrier backbones, L3VPNs, SFC, and 5G/6G network slicing |
| **Disaggregated Radio Access Network (Open RAN)** | O-RAN Alliance Architecture | Open interfaces (A1, O1, E2) linking SMO, Non-RT RIC, Near-RT RIC, and E2 Nodes | Virtualized RAN disaggregation (O-CU, O-DU, O-RU) with xApp control loops | Regional/metro cellular access; 10ms to 1s Near-RT RIC control loops |
| **Next-Generation Cellular Network (6G)** | ITU-R M.2160-0 (IMT-2030 Framework) | Centimeter-wave, millimeter-wave, and THz spectrum with native AI/ISAC | Integrated Sensing and Communication (ISAC) and ubiquitous intelligence | Global cellular/non-terrestrial coverage; 0.1 -- 1ms latency; $10^6$ – $10^8$ devices/km$^2$ |
| **Quantum Information Network** | IEEE / IETF Emerging | Quantum bits (qubits), | Entanglement swapping, quantum | Ultra-secure continental/satellite QKD and |

| Network Domain & Classification | Primary Standard / Framework | Core Transport & Signaling Mechanism | Key Architectural Innovation | Primary Operational Scope & Target Latency / Scale |
|---|---|---|---|---|
| **(Quantum Internet)** | Quantum Stacks | EPR entanglement distribution, and quantum repeaters | teleportation, and no-cloning security | distributed quantum computing |

## 2. Programmable Core and Transport Networks: Segment Routing over IPv6 (SRv6)

The architectural evolution of carrier-grade Wide Area Networks (WANs) and cloud-scale backbones has been defined by the effort to decouple path control from transit state maintenance. In legacy Multiprotocol Label Switching (MPLS) architectures, forwarding relies on locally significant label stacks ([label1, label2, label3] + IP payload) that require dedicated control-plane signaling protocols—specifically the Label Distribution Protocol (LDP) for shortest-path label binding and Resource Reservation Protocol-Traffic Engineering (RSVP-TE) for explicit path reservation. As network topologies scale to support cloud-to-edge distributed computing and fine-grained 5G/6G network slicing, maintaining per-tunnel RSVP-TE state across thousands of transit routers induces prohibitive control-plane overhead and convergence delays.

Segment Routing (SR), standardized in IETF RFC 8402, resolves this scalability barrier by adopting the source routing paradigm. Under SR, an ingress node steers a packet through an ordered list of topological or service instructions—termed "segments"—encoded directly into the packet header, entirely eliminating the need for transit path signaling protocols such as LDP or RSVP-TE. While early implementations mapped segments onto existing MPLS data planes (SR-MPLS), Segment Routing over IPv6 (SRv6) achieves a far deeper architectural unification by instantiating the segment routing paradigm natively within the IPv6 forwarding plane.

## 2.1 The SRv6 Data Plane and 128-Bit SID Structure

The SRv6 framework is anchored by two primary IETF standards: RFC 8754, which defines the IPv6 Segment Routing Header (SRH, designated as IPv6 Routing Extension Header Type 4), and RFC 8986, which establishes the SRv6 Network Programming model. Rather than treating an IPv6 address merely as a topological location identifier, RFC 8986 redefines the 128-bit IPv6 Segment Identifier (SID) as an executable network instruction composed of three distinct bit fields formatted as LOC:FUNCT:ARG.

Comprising the L most significant bits of the 128-bit SID (where L is configurable from 4 to 96 bits, and typically allocated as 48 bits in standard deployments), the Locator (LOC) serves as a routable IPv6 summary prefix advertised via Interior Gateway Protocols (IGPs) such as IS-IS or OSPF. The Locator is structured in the format B:N, where B represents a common SRv6 domain block address shared by all routers within the domain, and N represents a unique node-specific identifier. Because the Locator is globally routable within the underlay, any transit IPv6 router—even one completely unaware of SRv6—can forward the packet toward the segment endpoint node using standard Longest Prefix Match (LPM) lookups in its Forwarding Information Base (FIB). Immediately following the Locator, the Function (FUNCT) field (typically 16bits) specifies an opaque, locally significant instruction bound to the target segment endpoint node. When a packet arrives at the node identified by LOC, the Function bits instruct the router's forwarding engine to execute a specific forwarding, decapsulation, Virtual Routing and Forwarding (VRF) lookup, or service-chaining behavior. Occupying the remaining least significant bits of the 128-bit address, the optional Arguments (ARG) field carries flow-specific metadata, Quality of Service (QoS) parameters, or service arguments required by the executing Function (signaled in the control plane via extensions such as RFC 9819). For instance, in an operational SID formatted as 5f00:1:0:e001::, the 48-bit prefix 5f00:1::/48 serves as the Locator identifying router R1, e001 designates an End.X cross-connect function to a specific next-hop, and the trailing bits represent zeroed arguments.

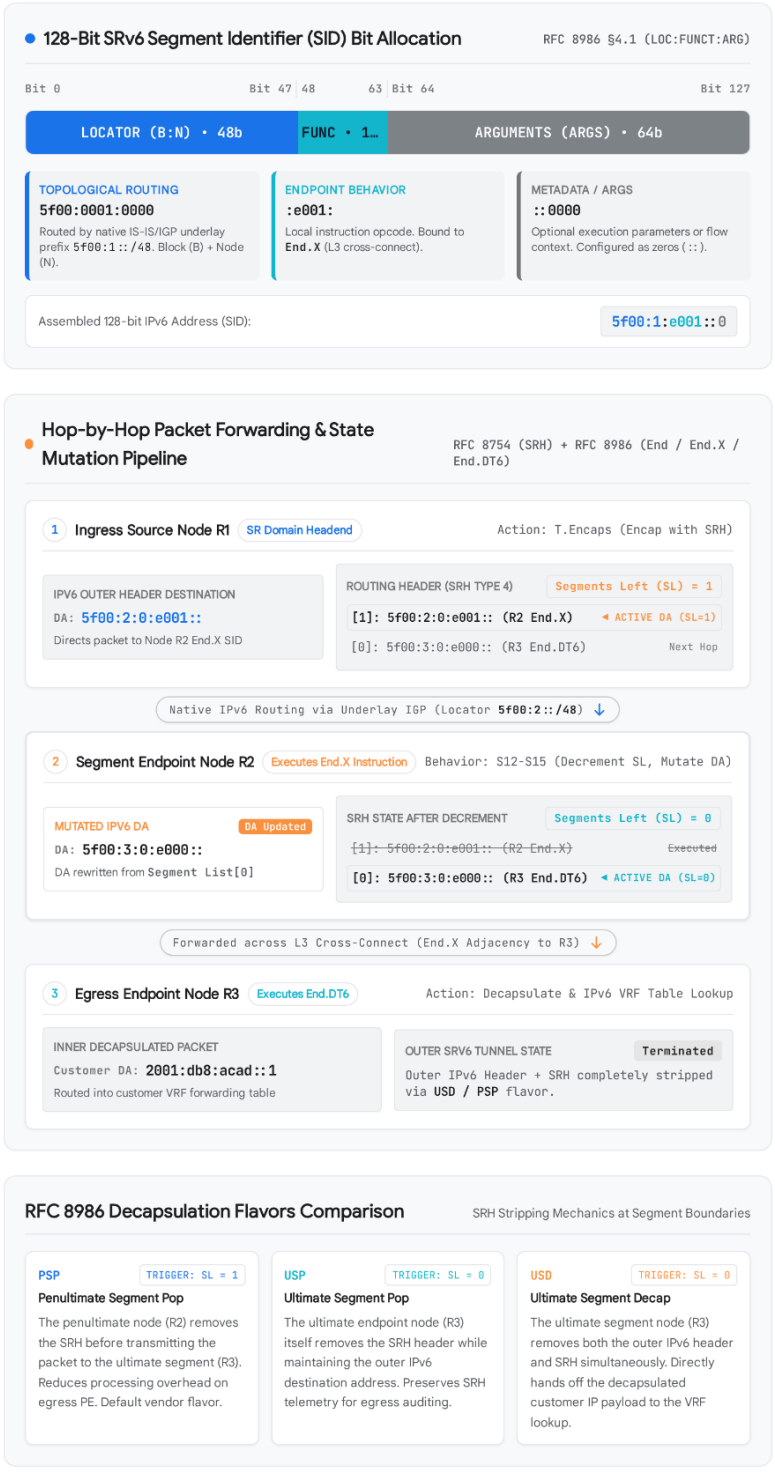



***Figure 1: SRv6 Encodes Distributed Network Programs Directly Within 128-Bit IPv6 Addresses***

Architectural decomposition of Segment Routing over IPv6 (SRv6) per IETF RFC 8754 and RFC 8986. Top: Bit-level allocation of a 128-bit SRv6 Segment Identifier (SID). Bottom: Hop-by-hop mutation of the IPv6 Destination Address (DA) and Segments Left (SL) pointer across a three-node topology executing End.X and End.DT6 behaviors, alongside SRH decapsulation flavors.

Data sources: Segment Routing Tech, Nokia Documentation, IETF RFC 8986, SRv6.md, OneUptime

## 2.2 Forwarding Mechanics, Endpoint Behaviors, and SRH Decapsulation Flavors

In an SRv6-enabled domain, nodes operate in one of three functional roles defined by RFC 8754: SR Source Nodes, Transit Nodes, and SR Segment Endpoint Nodes. An SR Source Node—which may be an originating host or an ingress Provider Edge (PE) router—steers traffic by either inserting an SRH into an existing IPv6 packet via transit behaviors (T.Insert or T.Insert.Red, where the reduced variant omits the final SID in the SRH because it is already carried in the outer IPv6 Destination Address) or encapsulating the packet within a new outer IPv6 header and optional SRH (T.Encaps). The SRH contains an ordered array of 128-bit SIDs indexed in reverse order of traversal, meaning Segment List[0] represents the ultimate destination segment and Segment List[Last Entry] represents the initial segment endpoint.

When the packet is transmitted by the SR Source Node (such as ingress router R1 forwarding across R2 to egress R3), the outer IPv6 Destination Address (DA) is populated with the first active segment (Segment List[Segments Left], such as R2's End.X SID 5f00:2:0:e001::), and the Segments Left (SL) field is initialized to match the index of that entry. Intermediate Transit Nodes whose local addresses do not match the outer IPv6 DA do not inspect or parse the SRH; they simply forward the packet at line rate based on standard IPv6 FIB lookups of the Locator prefix.

Only when the packet arrives at an SR Segment Endpoint Node—where the packet's IPv6 DA matches a locally instantiated SID in the router's FIB—is the SRH processed. For the foundational End behavior codified in RFC 8986 Section 4.1, the endpoint node executes a deterministic validation and forwarding sequence. Initially, the router evaluates whether Segments Left == 0; if this condition holds, SRH processing halts immediately and the forwarding engine proceeds to process the next header in the packet as identified by the Next Header field in the routing header. If segments remain, the node verifies whether the IPv6 Hop Limit <= 1, in which case it transmits an ICMPv6 Time Exceeded message (Code 0: Hop limit exceeded in transit) back to the Source Address, interrupts packet processing, and discards the packet. Next, the router validates header boundary integrity by computing max_LE = (Hdr Ext Len / 2) - 1. If (Last Entry > max_LE) or (Segments Left > Last Entry + 1), the node generates an ICMPv6 Parameter Problem message (Code 0: Erroneous header field encountered) with the pointer directed to the Segments Left field, halts processing, and drops the packet. Once header integrity is confirmed, the router decrements the IPv6 Hop Limit by 1, decrements Segments Left by 1, overwrites the outer IPv6 DA with Segment List[Segments Left], and submits the updated packet to the egress IPv6 FIB lookup within the same VRF or table as the ingress interface. When processing an upper-layer

header of a packet matching a locally instantiated End SID, the router checks whether the upper-layer header type is permitted by local configuration—such as allowing ICMPv6 (Type 58) so operators can ping active SIDs for Operations, Administration, and Maintenance (OAM)—and otherwise rejects the packet with an ICMPv6 Parameter Problem message (Code 4: SR Upper-layer Header Error) with the pointer set to the offset of the upper-layer header.

To optimize hardware processing at the network edge, RFC 8986 defines three SRH removal variations—known as flavors—that modify End, End.X, and End.T behaviors. The Penultimate Segment Pop (PSP) flavor triggers at the penultimate segment endpoint when the incoming packet has SL = 1 and the decremented Segments Left counter reaches 0. Under PSP, the penultimate router strips the SRH before forwarding the clean IPv6 packet to the final destination, offloading extension header removal from the egress PE router; consequently, PSP is the default and most widely deployed flavor across vendor implementations. Conversely, the Ultimate Segment Pop (USP) flavor leaves the SRH intact until the packet reaches the ultimate segment endpoint (SL = 0), where the final node removes the SRH before processing the next header—a capability required whenever the egress PE must inspect metadata, policy parameters, or OAM telemetry carried inside the SRH. Finally, the Ultimate Segment Decapsulation (USD) flavor also executes at the ultimate segment endpoint (SL = 0), removing the entire outer IPv6 header and its attached SRH to expose and forward the encapsulated inner payload.

**Table 2: Standardized SRv6 Endpoint Behaviors, Forwarding Semantics, and Operational Use Cases (RFC 8986)**

| SRv6 Behavior / Codepoint | Functional Role | Forwarding Plane Execution Semantics | Primary Deployment Use Case & Architectural Nuance |
|---|---|---|---|
| End | Prefix / Node Segment Endpoint | Decrements Hop Limit and SL, updates IPv6 DA with Segment List[SL], performs FIB lookup in ingress table. | Strict or loose topological traffic engineering via specific transit routers. |
| End.X | L3 Cross-Connect | Updates IPv6 DA from SRH and forwards directly | Fast Reroute (TI-LFA) and explicit |

| SRv6 Behavior / Codepoint | Functional Role | Forwarding Plane Execution Semantics | Primary Deployment Use Case & Architectural Nuance |
|---|---|---|---|
| | (Adjacency SID) | across a pre-bound Layer 3 point-to-point adjacency without FIB lookup. | link-level path steering. |
| LAN-End.X | Broadcast LAN Adjacency Endpoint | Equivalent to End.X, but bound to a specific next-hop neighbor over a multi-access broadcast LAN interface. | Explicit adjacency steering across shared Ethernet exchange segments. |
| End.T | Specific Table Lookup Endpoint | Updates IPv6 DA and performs the egress FIB lookup within a specific local routing table/VRF associated with the SID. | Multi-topology routing and virtualized core slicing. |
| End.DT4 | Decapsulation & IPv4 Table Lookup | Removes outer IPv6/SRH encapsulation and performs a prefix lookup in a specific IPv4 VRF or global IPv4 table. | Multi-homed Layer 3 VPN (VPN-IPv4) replacing per-VRF MPLS VPN labels. |
| End.DT6 | Decapsulation & IPv6 Table Lookup | Removes outer IPv6/SRH encapsulation and | Multi-homed Layer 3 VPN (VPN-IPv6) over |

| SRv6 Behavior / Codepoint | Functional Role | Forwarding Plane Execution Semantics | Primary Deployment Use Case & Architectural Nuance |
|---|---|---|---|
| | | performs a prefix lookup in a specific IPv6 VRF or global IPv6 table. | pure IPv6 backbones. |
| End.DT46 | Dual-Stack Decapsulation & Lookup | Decapsulates outer header and performs lookup for either inner IPv4 or IPv6 packets within a unified dual-stack VRF. | Conserves SID address space by replacing separate End.DT4 and End.DT6 SIDs. |
| End.DX4 / End.DX6 | Decapsulation & Direct L3 Cross-Connect | Strips outer IPv6/SRH header and cross-connects inner IPv4/IPv6 packet directly to a designated Customer Edge (CE) next-hop. | Single-homed per-CE direct VPN forwarding without VRF table lookup overhead. |
| End.AS / End.AD / End.AM / End.AN | Service Function Chaining (SFC) Proxy & Aware Behaviors | Steers traffic through SR-aware or SR-unaware middleboxes (via static, dynamic, or masquerading proxy encapsulation). | In-line Service Function Chaining across Firewalls, IDS, IPS, NAT, and DPI appliances. |

## 2.3 Control Plane Integration, Service Function Chaining, and 5G/6G Network Slicing

A critical second-order outcome of the SRv6 architecture is how it collapses the control plane across previously disjointed network services. In legacy architectures, deploying Layer 3 VPNs, Traffic Engineering, and Service Function Chaining (SFC, RFC 7665) required separate overlay and underlay protocols (such as MP-BGP, LDP, RSVP-TE, and Network Service Headers). Under SRv6, BGP extensions defined in RFC 9252 allow egress PE routers to advertise VPN prefixes alongside their corresponding End.DT4, End.DT6, or End.DT46 SIDs directly across domain boundaries, preserving QoS markings and VPN context without MPLS label translation.

Simultaneously, centralized and hybrid traffic engineering is orchestrated via an SRv6 Controller combining a Path Computation Element (PCE, RFC 4655) and a Classification Rule Controller. Topology state and Flex-Algorithm constraints are gathered into the Traffic Engineering Database (TED) via BGP Link-State (BGP-LS, RFC 9552 and RFC 9857), while candidate SR Policies (RFC 9256) are instantiated on headend nodes using PCEP extensions for SRv6 (RFC 5440 and RFC 9603) or BGP SR Policy advertisements (RFC 9830) and FlowSpec rules (RFC 8955 and RFC 8956). Additional ecosystem standards further automate provisioning and multicast delivery, including SRv6 SID addressing conventions (RFC 9602), Colored Prefix Routing (RFC 9723), Locator assignment via DHCPv6 (RFC 10038), Replication Segments (RFC 9524), Multicast-only Fast Reroute with TI-LFA (RFC 9860), and Point-to-Multipoint SR Policies (RFC 9960). When integrated with Service Function Chaining, an SRv6 Service Function Manager configures network function instances—such as Deep Packet Inspection (DPI), Intrusion Prevention Systems (IPS), Intrusion Detection Systems (IDS), Network Address Translation (NAT), and Firewalls—as addressable service segments (End.AN for SR-aware functions, or End.AS, End.AD, and End.AM proxies for SR-unaware appliances). By binding distinct SID lists and Flex-Algorithm locators to specific 5G and 6G service classes, operators construct deterministic network slices that route Ultra-Reliable Low-Latency Communications (URLLC) over latency-minimized paths and enhanced Mobile Broadband (eMBB) over throughput-optimized paths within a single converged IPv6 underlay.

# 3. High-Performance Access, Storage Fabrics, and Disaggregated Radio Networks

While SRv6 modernizes wide-area transport, severe performance bottlenecks have shifted to the network periphery—specifically wireless local access, data center storage interconnects, and cellular radio access networks. Each of these domains has undergone a parallel architectural redesign to eliminate protocol translation latency and static resource allocation.

## 3.1 Extremely High Throughput Wireless Local Area Networks (IEEE 802.11be / Wi-Fi 7)

In enterprise and industrial Wireless Local Area Networks (WLANs), the proliferation of immersive extended reality (XR), real-time industrial automation, and high-density client environments exposed the scheduling and channel-width limitations of IEEE 802.11ax (Wi-Fi 6/6E). Standardized in IEEE 802.11be-2024 and designated as Extremely High Throughput (EHT), Wi-Fi 7 fundamentally alters wireless Medium Access Control (MAC) and Physical Layer (PHY) architecture.

Whereas legacy Wi-Fi generations restricted client devices to transmitting and receiving over a single radio band at any given instant, IEEE 802.11be introduces Multi-Link Operation (MLO) at the MAC layer. MLO enables a single Wi-Fi 7 client and Access Point (such as enterprise multi-band platforms) to establish concurrent logical links across multiple frequency bands simultaneously. By aggregating spectrum dynamically and steering latency-sensitive frames onto whichever link exhibits the lowest instantaneous interference or contention, Wi-Fi 7 achieves both ultra-fast aggregate data transmission rates and bounded, deterministic low latency that approaches wired Ethernet reliability.

## 3.2 Lossless Storage Area Networks and NVMe over Fabrics (NVMe-oF)

Inside hyperscale data centers and AI training clusters, Storage Area Networks (SANs) have experienced an acute mismatch between solid-state media speed and network transport protocol overhead. Traditional SANs relied on Fibre Channel fabrics transporting Small Computer System Interface (SCSI) commands designed decades ago for spinning magnetic disks. When backend storage arrays transitioned to high-concurrency Non-Volatile Memory Express (NVMe) flash drives, legacy SAN architectures required protocol gateways to receive SCSI commands over Fibre Channel and translate and interpret them into native NVMe commands. This command translation and single-queue serialization introduced substantial CPU interrupt overhead and tail latency.

NVMe over Fabrics (NVMe-oF) resolves this bottleneck by extending the native NVMe submission and completion queue architecture end-to-end across the network fabric without SCSI translation. In production data centers, NVMe-oF operates across three primary transport bindings tailored to distinct infrastructure priorities. The highest-performance implementation, NVMe-oF over RDMA (including RoCE, or RDMA over Converged Ethernet), leverages Remote Direct Memory Access to bypass the host operating system kernel and TCP/IP stack entirely, transferring NVMe capsules directly between host memory and storage target memory via lossless Ethernet or InfiniBand hardware offloads. For enterprises with existing dedicated storage switchgear, NVMe over Fibre Channel (FC-NVMe) preserves the physical Fibre Channel fabric while replacing the upper-layer SCSI command set with native encapsulated NVMe commands, thereby eliminating translation latency. Alternatively, NVMe over TCP

(NVMe/TCP) encapsulates NVMe Protocol Data Units (PDUs) within standard TCP/IP datagrams, trading a modest latency penalty relative to RDMA/RoCE for universal interoperability across standard data center Ethernet fabrics.

### 3.3 Disaggregated Cellular Architectures: Open RAN (O-RAN) Control Loops

In the cellular access domain, traditional Radio Access Networks (RANs) were deployed as monolithic, vertically integrated systems where the radio hardware, baseband processing, and control software were tightly coupled through proprietary vendor interfaces. The O-RAN Alliance architecture dismantles this vendor lock-in by disaggregating the base station into standardized functional components—the O-RAN Central Unit (O-CU), O-RAN Distributed Unit (O-DU), O-RAN Radio Unit (O-RU), and legacy next-generation O-eNB—collectively managed as E2 Nodes under an open, cloud-native intelligence hierarchy.

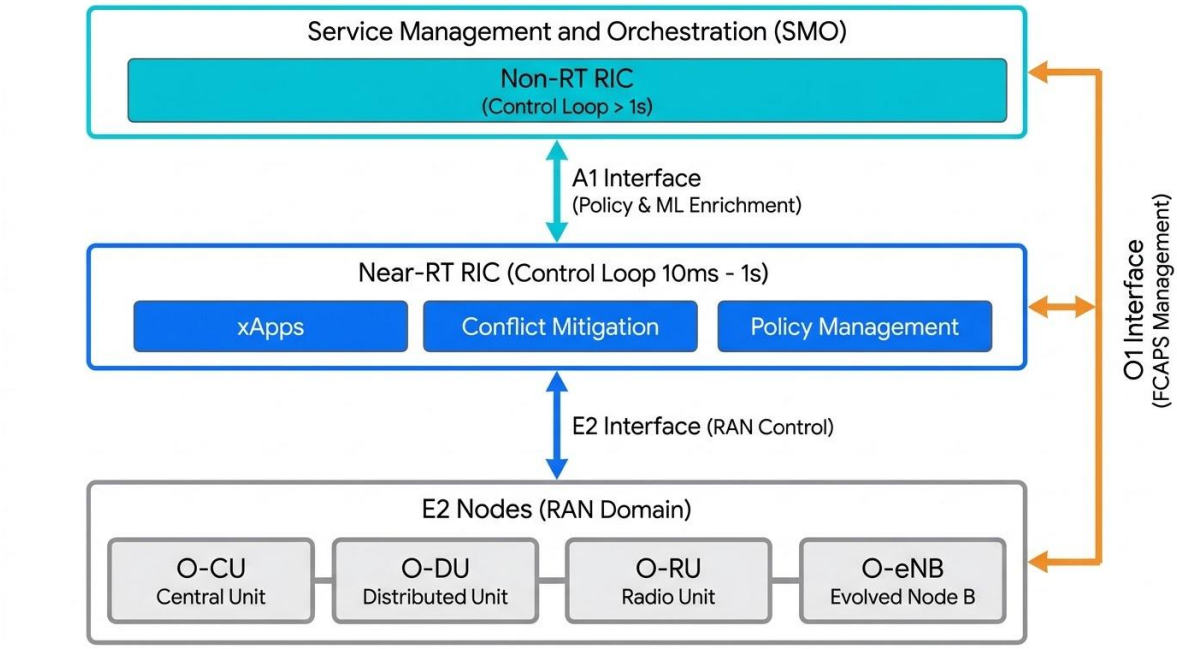



***Figure 2: O-RAN Disaggregates Cellular Base Stations Into Multi-Timescale Intelligent Control Loops***

High-level architecture of an Open Radio Access Network (O-RAN). The Service Management and Orchestration (SMO) framework houses the Non-RT RIC (operating at intervals greater than 1 second) and communicates with the Near-RT RIC (operating on 10 ms to 1 s control loops via xApps) over the A1 interface for policy and ML management, while the O1 interface provides FCAPS management across all O-RAN components and the E2 interface governs E2 Nodes (O-CU, O-DU, O-RU, and O-eNB).

Within the O-RAN structural model, network intelligence is partitioned across two distinct RAN Intelligent Controllers (RICs) linked by three newly standardized open interfaces: A1, O1, and E2. Situated at the top of the management hierarchy, the Service

Management and Orchestration (SMO) framework provides domain-wide Fault, Configuration, Accounting, Performance, and Security (FCAPS) management to all O-RAN network functions via the O1 interface. Embedded within the SMO is the Non-Real-Time RIC (Non-RT RIC), which executes non-real-time radio resource management, long-term traffic trend analysis, and machine learning (ML) model training over control intervals exceeding 1second. The Non-RT RIC connects directly to the Near-Real-Time RIC (Near-RT RIC) over the A1 interface. Rather than carrying raw user-plane traffic, the A1 interface transmits declarative intent policies, external enrichment information, and trained ML model updates down to the edge. Deployed at the regional or metro edge, the Near-RT RIC executes use-case-specific closed control loops operating within strict 10ms to 1s boundaries. It hosts modular third-party microservice applications known as xApps, along with internal routing and topology services, policy management, messaging infrastructure, and a conflict mitigation engine. Conflict mitigation is essential in disaggregated networks because two independent xApps—for instance, one optimizing mobility handover and another minimizing base station energy consumption—can issue contradictory parameter adjustments; the Near-RT RIC resolves these conflicts before transforming the approved directives into node-specific configurations installed over the E2 interface onto the underlying E2 Nodes (O-CU, O-DU, O-RU, and O-eNB), which execute real-time radio resource management and mobility coordination.

## 4. Next-Generation Ubiquitous Connectivity: The IMT-2030 (6G) Framework

While 5G (IMT-2020) established the foundation for service-based core architectures and initial network slicing, emerging applications envisioned for the 2030 decade—such as high-fidelity mobile holograms, immersive mixed reality, autonomous swarm robotics, and real-time digital replication (digital twins)—demand deterministic throughput, sub-millisecond latency, and spatial sensing capabilities that exceed the physical and architectural ceilings of 5G. To standardize this transition, the International Telecommunication Union Radiocommunication Sector (ITU-R) Working Party 5D finalized Recommendation ITU-R M.2160-0 in November 2023, establishing the formal IMT-2030 Framework for sixth-generation (6G) mobile systems in alignment with the Next Generation Mobile Networks (NGMN) Alliance and 3GPP standardization roadmaps.

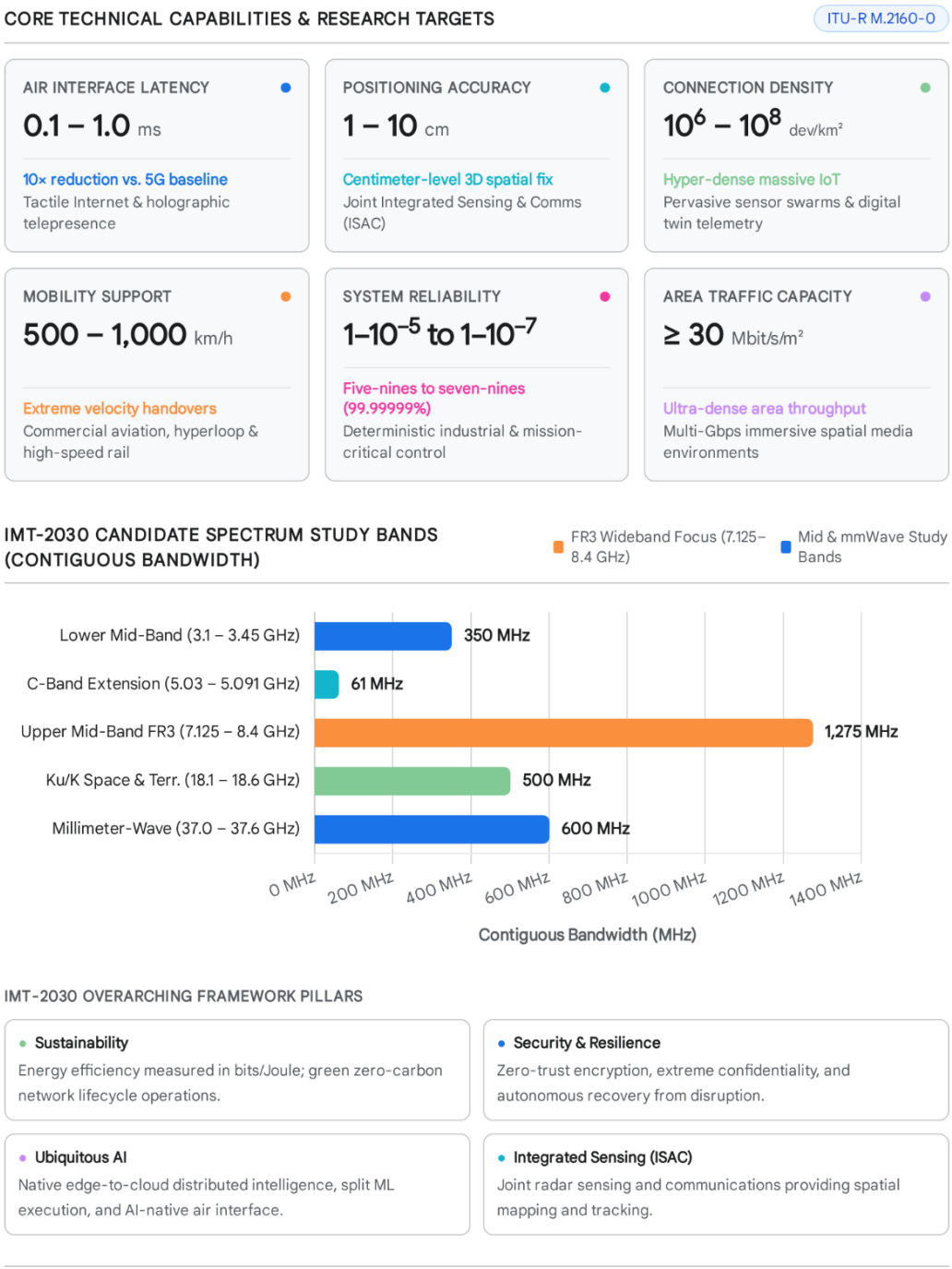



***Figure 3: IMT-2030 Establishes Sub-Millisecond Latency and Centimeter Positioning Targets for 6G***

Target capability ranges and candidate spectrum study bands for sixth-generation (6G) networks under the ITU-R IMT-2030 framework (Recommendation M.2160-0). Top: Quantitative research targets across six core operational dimensions. Bottom: Five primary frequency bands designated for 6G propagation, coexistence, and system study across centimeter-wave and millimeter-wave regimes.

Data sources: NGMN Alliance (ITU-R Recommendation M.2160-0 Framework), 5G Americas (ITU's IMT-2030 Vision: Navigating Towards 6G)

## 4.1 Quantitative Performance Envelopes and Spectrum Implications

The IMT-2030 specification expands upon classical cellular metrics by pushing physical-layer boundaries while introducing entirely new functional capabilities. At the transport level, 6G systems target air-interface latency between 0.1 and 1ms, coupled with frame delivery reliability ranging from $1\text{-}10^{-5}$ (99.999%) to $1\text{-}10^{-7}$ (99.99999%). To support hyper-dense autonomous environments and industrial sensor grids, connection density targets scale to between $10^6$ and $10^8$ active devices/km$^2$, while area traffic capacity is projected to reach 30 Mbit/s/m$^2$ and beyond. Simultaneously, Doppler compensation and waveform design must sustain seamless handovers at mobility velocities between 500 and 1,000 km/h to serve high-speed rail and aviation platforms.

Achieving these performance envelopes requires a strategic multi-band spectrum architecture. While sub-Terahertz (THz) spectrum offers vast contiguous bandwidths for ultra-short-range terabit links, propagation path loss and site densification costs necessitate a balanced spectrum portfolio. Consequently, regulatory and industry studies have prioritized five specific frequency bands for coexistence and propagation investigation: 3.1 -- 3.45GHz (350MHz width), 5.03 -- 5.091GHz (61MHz width), the upper mid-band ("FR3") at 7.125 -- 8.4 GHz (1,275MHz width), 18.1 -- 18.6GHz (500MHz width), and millimeter-wave spectrum at 37.0 -- 37.6GHz (600MHz width). The 7.125 -- 8.4 GHz band is particularly pivotal because it offers over 1.2GHz of spectrum capable of supporting wide-channel massive MIMO arrays while retaining macro-cell propagation characteristics superior to millimeter-wave frequencies.

## 4.2 Native AI, Integrated Sensing and Communication (ISAC), and Sustainability

Beyond raw throughput and latency, IMT-2030 and NGMN design guidelines fundamentally redefine the cellular network from a pure bit-pipe into a distributed sensory and computational fabric. Through Integrated Sensing and Communication (ISAC), 6G networks utilize high-frequency radio waveforms and massive antenna arrays as bistatic and monostatic radar transceivers, extracting range, Doppler velocity, and angle-of-arrival reflections from passive objects in the surrounding environment. This native sensing capability enables 1 to 10cm spatial positioning accuracy, real-time environmental mapping, and gesture or posture recognition without requiring target objects to carry active transponders. Simultaneously, rather than treating Artificial Intelligence merely as an over-the-top application, IMT-2030 embeds ubiquitous intelligence and distributed computing for AI directly into the radio and core architecture. This architectural convergence flattens the computation-communication gap, delivering seamless edge-to-cloud connectivity across disaggregated, neutral-host, and shared network infrastructures while supporting digital inclusion across unconnected regions. Recognizing the severe environmental, energy, and economic trade-offs associated with site densification, both ITU-R M.2160-0 and NGMN elevate environmental sustainability to a primary architectural requirement, evaluating network

energy efficiency in terms of transmitted bits per Joule (bit/J) alongside overarching guarantees of security, trustworthiness, and rapid recovery from disruptive events.

## 5. Cryptographic Hardening of Classical Networks: Post-Quantum Cryptography (PQC)

As classical networks evolve toward SRv6 backbones and 6G wireless fabrics, their underlying security protocols—including Transport Layer Security (TLS), IP Security (IPsec), Secure Shell (SSH), and Public Key Infrastructure (PKI)—face an existential cryptographic transition. Classical public-key cryptography relies on the computational intractability of integer factorization (RSA) or the discrete logarithm problem over finite fields and elliptic curves (Diffie-Hellman, ECDH, DSA, ECDSA, and Ed25519). A sufficiently powerful quantum computer running Shor's algorithm solves both integer factorization and discrete logarithms in polynomial time, providing an exponential speedup that completely compromises all deployed classical asymmetric cryptosystems.

Conversely, symmetric-key cryptographic algorithms (such as AES-256) and cryptographic hash functions are not broken by Shor's algorithm. Against symmetric primitives, Grover's quantum search algorithm yields at most a quadratic speedup on brute-force key search, reducing the effective security strength of AES-256 to roughly 128 bits—which remains computationally sound. Consequently, the post-quantum migration scope is strictly focused on replacing asymmetric primitives: key exchange/encapsulation, digital signatures, and public-key encryption. Because threat actors actively engage in Harvest Now, Decrypt Later (HNDL) espionage—capturing encrypted TLS and VPN network traffic today to decrypt once cryptographically relevant quantum computers mature—the deployment of Post-Quantum Cryptography (PQC) across classical network protocols is an immediate operational imperative.

### 5.1 Finalized NIST Post-Quantum Cryptographic Standards (FIPS 203, FIPS 204, and FIPS 205)

Following a multi-year global cryptanalysis competition initiated in 2016 with finalists selected in 2022, the U.S. National Institute of Standards and Technology (NIST) formally released the first three finalized Post-Quantum Cryptography Federal Information Processing Standards on August 13, 2024 (effective August 14, 2024): FIPS 203, FIPS 204, and FIPS 205.

FIPS 203 specifies the Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM), derived from the CRYSTALS-Kyber algorithm and grounded in the computational hardness of the Module Learning With Errors (MLWE) problem over structured polynomial matrices and rings. Unlike classical Diffie-Hellman or ECDH mutual key exchanges, ML-KEM operates via encapsulation and decapsulation: one party generates a public/private key pair, the communicating peer uses the recipient's

public key to encapsulate a shared secret into a ciphertext over the insecure channel, and the recipient decapsulates the ciphertext to establish the shared secret key. FIPS 203 incorporates cryptographic domain separation during key generation to prevent cross-protocol attacks and defines three parameter sets in order of increasing security strength and decreasing performance: ML-KEM-512 (providing baseline security), ML-KEM-768 (the standard deployment target for civilian TLS and VPN systems), and ML-KEM-1024 (NIST Security Level 5, mandated by the Commercial National Security Algorithm Suite 2.0 [CNSA 2.0] for classified and national security systems).

Complementing key establishment, FIPS 204 standardizes the Module-Lattice-Based Digital Signature Algorithm (ML-DSA), which also relies on the structured lattice MLWE problem to replace RSA and ECDSA for digital authentication, PKI certificate signing, and software or firmware code signing. In enterprise deployments, ML-DSA-65 serves as the standard commercial target, whereas ML-DSA-87 fulfills high-security CNSA 2.0 requirements. As a structurally independent alternative, FIPS 205 specifies the Stateless Hash-Based Digital Signature Standard (SLH-DSA), derived from SPHINCS+ and relying exclusively on hash function security without any lattice assumptions. Internally, SLH-DSA generates signatures using a Winternitz One-Time Signature Plus (WOTS+) scheme at the leaf nodes, authenticates those WOTS+ public keys via a stacked multi-layer Merkle tree (hypertree) to keep tree heights manageable, and executes top-level signing via a Forest of Random Subsets (FORS) few-time signature scheme. Because SLH-DSA is computationally slower and produces substantially larger signatures, it is reserved as a conservative backup for long-lived root Certificate Authority (CA) keys and high-assurance code signing where mathematical diversification away from lattices is prioritized over throughput. To further expand cryptographic diversity, NIST selected HQC in March 2025 as a code-based backup KEM under NIST IR 8545, released supplemental KEM guidance under SP 800-227, and continues developing a fourth signature standard based on the FALCON algorithm—designated FN-DSA (draft FIPS 206)—expected between late 2026 and early 2027.

**Table 3: Comparative Analysis of Finalized NIST Post-Quantum Cryptography Standards (FIPS 203, FIPS 204, and FIPS 205)**

| Cryptographic Attribute | FIPS 203 (ML-KEM) | FIPS 204 (ML-DSA) | FIPS 205 (SLH-DSA) |
|---|---|---|---|
| **Primary Cryptographic Function** | Key Encapsulation Mechanism (shared secret establishment) | General-purpose Digital Signatures (authentication & non-repudiation) | Stateless Backup Digital Signatures (conservative |

| Cryptographic Attribute | FIPS 203 (ML-KEM) | FIPS 204 (ML-DSA) | FIPS 205 (SLH-DSA) |
|---|---|---|---|
| | | | root authentication) |
| **Underlying Mathematical Hardness** | Structured Lattice: Module Learning With Errors (MLWE) | Structured Lattice: Module Learning With Errors (MLWE) | Hash-Function Security Only (WOTS+, Merkle Hypertrees, FORS) |
| **Classical Primitives Replaced** | RSA Key Transport, Diffie-Hellman (DH), ECDH | RSA Signatures, DSA, ECDSA, Ed25519 | RSA and ECDSA Signatures (specifically as a non-lattice fallback) |
| **Standardized Parameter Sets** | ML-KEM-512, ML-KEM-768 (civilian default), ML-KEM-1024 (CNSA 2.0 Level 5) | Includes ML-DSA-65 (civilian default) and ML-DSA-87 (CNSA 2.0) | Multiple parameter sets optimized for small size vs. fast computation |
| **Key / Ciphertext / Signature Size vs. ECC** | ~ 15 × to 25 × larger than classical ECDH | ~ 15 × to 50 × larger than classical ECDSA | ~ 100 × to 700 × larger than ECDSA (7,856 to 49,856 bytes) |
| **Computational Speed Profile** | Fast; comparable to or faster than classical KEMs on modern CPUs | Fast; suitable for high-volume TLS and code-signing workloads | Slow; signature generation requires hundreds of milliseconds per operation |

| Cryptographic Attribute | FIPS 203 (ML-KEM) | FIPS 204 (ML-DSA) | FIPS 205 (SLH-DSA) |
|---|---|---|---|
| **Primary Network & Systems Fit** | TLS 1.3 handshakes, IPsec/IKEv2 VPNs, SSH key exchange | TLS leaf/intermediate certificates, PKI, software/firmware code signing | Long-lived Root CA certificates and ultra-conservative firmware roots |

## 5.2 Protocol Engineering and Enterprise Migration Sequencing

Deploying PQC across global IP networks introduces profound second- and third-order engineering challenges that extend well beyond swapping software libraries. As quantified in Table 3, ML-KEM keys and ciphertexts are 15 to 25 × larger than ECDH, ML-DSA signatures are 15 to 50 × larger than ECDSA, and SLH-DSA signatures expand to between 7,856 and 49,856 bytes (100 to 700 × larger than ECDSA). In network protocols such as TLS and IPsec, embedding full certificate chains signed with ML-DSA or SLH-DSA causes handshake payloads to bloat substantially, requiring Content Delivery Networks (CDNs), Web Application Firewalls (WAFs), and middleboxes to support larger certificate structures without dropping fragmented packets.

Consequently, enterprise and carrier PQC migration must follow a strict operational dependency sequence. Implementation begins by compiling a comprehensive Cryptographic Bill of Materials (CBOM) to inventory every algorithm, key size, certificate, and protocol dependency across TLS endpoints, VPN gateways, PKI hierarchies, Hardware Security Modules (HSMs), and code-signing pipelines. Empirical enterprise audits reveal that organizations typically discover three to five times more cryptographic dependencies than initial estimates suggest, making automated discovery the primary prerequisite before algorithm replacement. Once vulnerable asymmetric instances are mapped, organizations prioritize deploying ML-KEM in hybrid mode (combining classical key exchange with ML-KEM) across TLS and VPN links. Hybrid key exchange immediately neutralizes Harvest Now, Decrypt Later interception while providing a cryptographic safety net if early PQC implementations contain undiscovered flaws, although it introduces the operational cost of a second migration later to pure PQC. Finally, digital signature migration across PKI hierarchies must proceed root-first and is gated by ecosystem readiness, including broad client verification support and FIPS 140-3 module validation through the NIST Cryptographic Module Validation Program (CMVP) across HSMs and cloud Key

Management Systems (KMS) throughout 2025–2027, culminating in full pure-PQC compliance under CNSA 2.0 and PCI DSS 4.0 timelines by 2033–2035.

## 6. The Quantum Internet: Physical Principles, Repeater Architectures, and 7G Horizons

While Post-Quantum Cryptography hardens classical networks using mathematical problems intractable for quantum computers, the Quantum Internet (QI) represents a fundamental paradigm shift in the physics of communication itself. Rather than transmitting classical bits ("0" or "1") via macroscopic electromagnetic pulses, the Quantum Internet transmits and distributes quantum bits (qubits)—such as the polarization state of a single photon or the spin state of an atomic system—across remote quantum nodes interconnected via fiber-optic or free-space channels.

### 6.1 Quantum Mechanical Foundations: Superposition, Entanglement, and No-Cloning

The architectural capabilities and physical constraints of the Quantum Internet derive directly from three foundational laws of quantum mechanics. Unlike a classical bit that occupies a definite state of either 0 or 1 like a coin landing on heads or tails, a single pure qubit state $|\psi\rangle$ can exist in a coherent linear superposition of basis states $|0\rangle$ and $|1\rangle$, expressed mathematically as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $\alpha$ and $\beta$ are complex numbers representing probability amplitudes. When a physical measurement is performed on the qubit $|\psi\rangle$, the superposition collapses probabilistically such that $|\alpha|^2$ represents the probability of obtaining $|\psi\rangle = |0\rangle$ and $|\beta|^2$ represents the probability of obtaining $|\psi\rangle = |1\rangle$, subject to the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1$$

Beyond single-qubit superposition, quantum entanglement—first recognized by Einstein, Podolsky, and Rosen (EPR) in 1935—binds two or more qubits into a correlated multipartite quantum state (such as a joint post-measurement state of $|00\rangle$ and $|11\rangle$). When a measurement is performed on qubit A (for example, along a $+45^\circ$ spin direction), the correlated outcome is instantaneously reflected at remote qubit B, enabling non-local information processing that transcends classical physics. Simultaneously, the no-cloning theorem dictates that an unknown quantum state cannot be deterministically copied or amplified without disturbing its state. This physical law creates a dual architectural effect: it guarantees intrinsic physical security against eavesdropping, yet it completely precludes the use of classical opto-electronic amplifiers or regenerative repeaters to boost attenuated quantum signals across long distances.

## 6.2 Quantum Repeaters, Entanglement Swapping, and Quantum Teleportation

Because photon loss scales exponentially with optical fiber length and the no-cloning theorem prohibits signal copying, achieving long-distance quantum communication requires dividing a long communication channel into shorter elementary segments bridged by Quantum Repeaters.

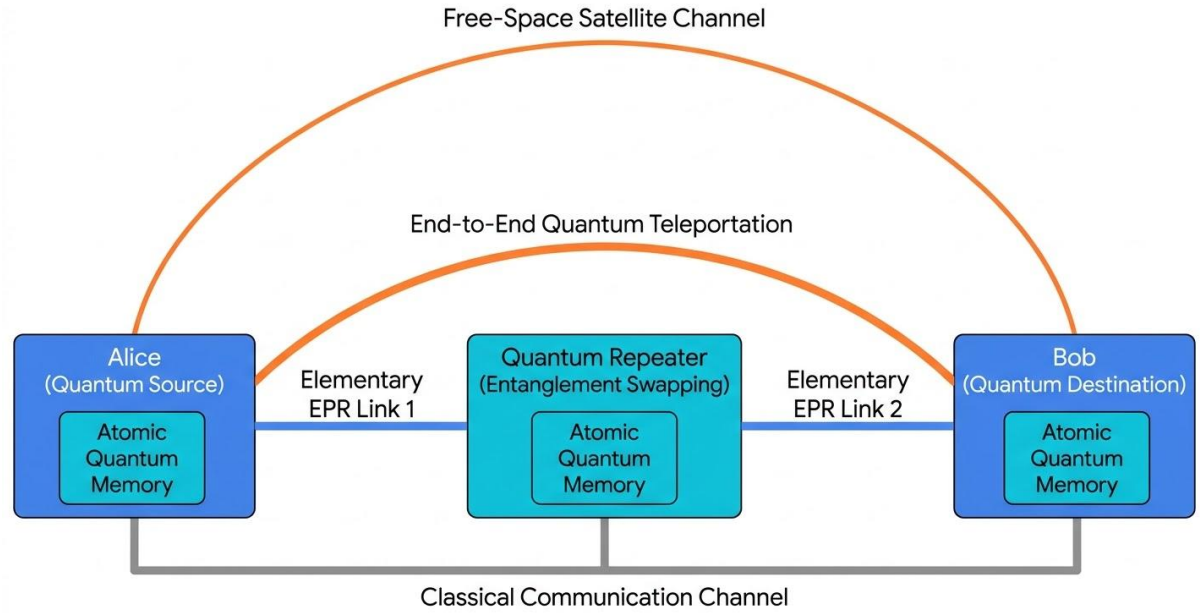



***Figure 4: Quantum Repeaters Overcome the No-Cloning Limit via Entanglement Swapping***

End-to-end architecture of long-distance quantum communication across a Quantum Repeater chain. Elementary entangled photon pairs (EPR pairs) are generated across short fiber segments and stored in stationary atomic qubit memories. Intermediate Quantum Repeaters execute entanglement swapping (Bell-state measurement) to establish end-to-end entanglement between Alice (Source) and Bob (Destination), enabling the transmission of an unknown quantum state via Quantum Teleportation alongside a classical signaling channel.

Long-distance quantum state transport across a repeater chain operates through the coordinated interplay of quantum memory storage, entanglement swapping, and quantum teleportation. Initially, entangled EPR pairs are distributed across adjacent short elementary fiber segments and stored within stationary atomic qubits acting as quantum memories at the nodes, which must maintain quantum coherence while neighboring segments establish entanglement. Once adjacent segments hold entangled pairs, the intermediate quantum repeater executes entanglement swapping—performing a local Bell-state measurement on its two locally stored half-pairs to splice the two short-range entanglements into a single end-to-end entangled EPR link between the remote source and destination nodes. To mitigate imperfect local operations and channel

decoherence, advanced architectures employ preventive quantum error coding and routing protocols such as DFER to distribute fidelity-guaranteed entangled pairs with high throughput. Once end-to-end entanglement is established between source and destination, an unknown quantum state $|\psi\rangle$ is transmitted over long distances via quantum teleportation using the shared EPR pair and local operations assisted by classical communications. Because the sender's local measurement outcomes must be communicated over a classical channel before the receiver can reconstruct the quantum state, the Quantum Internet fundamentally operates in tandem with classical communication networks.

## 6.3 The Four Pillars of the Quantum Ecosystem, Global Deployments, and 7G Horizons

Recent literature categorizes the architectural ecosystem of the Quantum Internet into four core domains: quantum computers, quantum networks, quantum cryptography, and quantum machine learning. Originally proposed to simulate quantum physics and subsequently generalized to encompass universal quantum computation beyond classical Turing machines, quantum computers hold disruptive potential for understanding molecular photosynthesis, designing catalysts for renewable fuels, and accelerating complex artificial intelligence systems. Cloud service providers including Microsoft Azure and Amazon Braket have already begun offering cloud-accessible quantum computing services, while global quantum investments approached $25 billion by mid-2021 alongside industry projections that 40% of large enterprises would integrate quantum computing into their workflows by 2025 despite a decade-long hardware maturation horizon.

Interconnecting these processors via wired fiber and wireless satellite quantum networks enables distributed quantum computation and quantum-enhanced sensing for healthcare and fundamental physics. In the security domain, quantum cryptography leverages Quantum Key Distribution (QKD) to establish cryptographic channels as secure as the information-theoretic One-Time-Pad (OTP). Recent breakthroughs demonstrate multi-stage repeater chains achieving trust-free continental QKD without overly demanding device specifications, alongside Deterministic Industrial IoT (Det-IIoT) modules embedded within orbital satellites to harden Space-Ground QKD against classical side-channel cyber-attacks. Furthermore, emerging quantum application protocols include quantum voting and survey systems that guarantee voter anonymity via quantum superposition while preventing double-voting through entanglement. In the computational intelligence domain, quantum machine learning utilizes paradigms such as quantum annealing to solve hyper-scale optimization and learning tasks. Large-scale national deployments are actively advancing globally, exemplified in India by the National Mission on Quantum Technologies and Applications (NM-QTA), the Quantum-Enabled Science & Technology (QuEST) program, and satellite-based quantum communication initiatives led by the Indian Space Research Organisation (ISRO). Looking toward 7G and beyond, researchers are designing a scalable, quantum-

compliant Open Systems Interconnection (OSI) framework featuring hierarchical quantum backbones, mobile quantum networks, noise-resilient Quantum Integrated Sensing and Communication (QISAC) co-design, programmable quantum stacks driven by state-machine RuleSets, digital twins, and AI-defined QNet agents.

**Table 4: Paradigm Comparison Between the Classical Internet and the Quantum Internet**

| Architectural Dimension | Classical Internet (Including SRv6 & 6G) | Quantum Internet (QI & 7G Quantum Stacks) |
|---|---|---|
| **Fundamental Information Unit** | Classical Bit (0 or 1 mutually exclusive logical states) | Quantum Bit / Qubit ($ |
| **Physical Signal Carrier** | Macroscopic electromagnetic waves, electrical voltages, or multi-photon optical pulses | Single photons (polarization/time-bin) and stationary atomic/spin qubits |
| **Signal Duplication & Long-Haul Extension** | Deterministic copying, regeneration, and opto-electronic amplification | Copying strictly forbidden by the **No-Cloning Theorem**; extended via **Quantum Repeaters** and **Entanglement Swapping** |
| **Data Transfer Mechanism** | Direct packet forwarding and header routing (e.g., IPv6 / SRv6 FIB lookup) | **Quantum Teleportation** via pre-distributed EPR entanglement + classical bit signaling |
| **Cryptographic Security Foundation** | Computational hardness assumptions (Structured Lattices / Hash functions via NIST FIPS 203/204/205) | Laws of quantum mechanics (No-cloning & entanglement collapse via QKD; One-Time-Pad equivalence) |

| Architectural Dimension | Classical Internet (Including SRv6 & 6G) | Quantum Internet (QI & 7G Quantum Stacks) |
|---|---|---|
| **Interdependency Between Planes** | Operates autonomously on classical forwarding, control, and management planes | Symbiotic: Requires a classical communication channel alongside the quantum channel to complete teleportation and QKD reconciliation |

## 7. Synthesis and Architectural Conclusions

A rigorous cross-layer synthesis of contemporary and emerging communication networks reveals three overarching architectural trajectories that will govern global infrastructure engineering over the next two decades. First, classical networking is converging onto stateless, header-programmed forwarding planes and disaggregated intelligent control loops. In wide-area transport, Segment Routing over IPv6 (SRv6, RFC 8986) eliminates the multi-protocol complexity of MPLS (LDP and RSVP-TE) by encoding topological paths, Layer 3 VPN table lookups (End.DT4, End.DT6, and End.DT46), and Service Function Chaining (End.AS, End.AD, End.AM, and End.AN) directly into 128-bit IPv6 Segment Identifiers. This structural simplification at the core is mirrored at the periphery by NVMe over Fabrics (NVMe-oF), which eliminates SCSI translation overhead across RDMA/RoCE data center fabrics, IEEE 802.11be (Wi-Fi 7), which introduces deterministic Multi-Link Operation across unlicensed bands, and Open RAN (O-RAN), which disaggregates monolithic base stations into O-CU, O-DU, and O-RU nodes governed by multi-timescale A1, O1, and E2 control loops.

Second, the boundary between communication, sensing, and computation is dissolving under the IMT-2030 (6G) framework. By targeting 0.1 to 1ms air-interface latency, $10^6$ to $10^8$ devices/km$^2$ connection density, 1-$10^{-5}$ to 1-$10^{-7}$ reliability, and 1 to 10cm 3D positioning across newly studied centimeter-wave (7.125 -- 8.4 GHz) and millimeter-wave (37.0 -- 37.6 GHz) bands, 6G transforms the wireless interface into an Integrated Sensing and Communication (ISAC) radar and distributed AI execution substrate constrained by strict bits-per-Joule sustainability budgets.

Third, quantum physics is simultaneously disrupting classical network cryptography and birthing a complementary quantum communication plane. Because Shor's algorithm renders RSA and elliptic curve cryptography obsolete, classical networks must execute an immediate, methodical transition to NIST's finalized Post-Quantum Cryptography standards—deploying ML-KEM (FIPS 203) in hybrid mode to halt Harvest Now, Decrypt Later attacks, while re-architecting PKI hierarchies, packet sizes,

and FIPS 140-3 HSMs to accommodate the 15 × to 700 × larger signature footprints of ML-DSA (FIPS 204) and SLH-DSA (FIPS 205) ahead of 2033–2035 CNSA 2.0 deadlines. Ultimately, the realization of the Quantum Internet will not replace the classical Internet; because quantum teleportation and entanglement swapping inherently depend on classical signaling channels to transmit Bell-state measurement outcomes, the future global information infrastructure will operate as a hybrid classical-quantum continuum. Within this dual-plane architecture, SRv6 and 6G/7G classical fabrics will transport high-volume deterministic data and control telemetry, while quantum repeater chains and satellite QKD constellations distribute multipartite entanglement for unbreakable cryptographic security and planetary-scale distributed quantum computing.

## 8. Declarations

### Author Contributions (CRediT Authorship Statement)

**SAMUELSON G:** Conceptualization, Methodology, Formal Analysis, Investigation, Technical Taxonomy Curation, Visualization (architectural schematics and comparative tables), Writing – Original Draft, Writing – Review & Editing, and Project Administration. The author has read and approved the final manuscript for publication.

### Competing Interests

No financial or non-financial competing interests exist in relation to the architectural frameworks, standards bodies, or commercial technologies analyzed in this study.

### Ethics Approval and Consent to Participate

Not applicable. This research consists exclusively of theoretical systems analysis, network protocol evaluation, and telecommunications standards synthesis. It does not involve human participants, human data, human tissue, or animal subjects; therefore, Institutional Review Board (IRB) or ethics committee approval was not required.

### Funding and Standards Compliance

This research received no specific grant from any commercial, public, or not-for-profit funding agency. All architectural specifications, protocol behaviors, and cryptographic parameters conform strictly to published IETF, IEEE, ITU-R, O-RAN Alliance, and NIST Federal Information Processing Standards.

### Data and Standards Availability

All technical data, RFC state-machine specifications, frequency allocations, and cryptographic parameter benchmarks analyzed in this study are publicly accessible through the respective standards development organizations (IETF, ITU-R, IEEE, NIST, and O-RAN Alliance) cited throughout the text and listed in the References section.

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